Mini Project: Cohort 21: Affordable Housing Price Estimator

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Frontend Module Streamlit Web Application

The frontend of the **House Price Prediction System** is developed using Streamlit, a Python framework that allows developers to create interactive web applications easily. Instead of building a complex website using HTML, CSS, and JavaScript, Streamlit enables the entire interface to be developed using Python. This makes it especially suitable for machine learning projects, where the main goal is to allow users to interact with a trained model in a simple and efficient way. The interface acts as a bridge between the user and the prediction system, making the application accessible even to non-technical users.

Libraries and Dependencies

To make the application functional, several Python libraries are used. Streamlit is responsible for designing the layout and interactive components of the webpage. Pandas is used to read and manage the dataset, particularly for retrieving valid location names. NumPy helps in converting user inputs into numerical form so that the model can understand them. The pickle library is used to load the previously trained machine learning model from a file. Together, these libraries ensure that the interface, data processing, and prediction system work smoothly.

Model Loading

The trained Linear Regression model is saved as a file after the training phase. In the frontend, this saved model is loaded so that it can be used to make predictions in real time. This approach saves time and computational resources because the model does not need to be trained again whenever the application runs. Once loaded, the model is ready to accept new inputs from users and estimate house prices based on patterns it learned from the training data.

Dataset Integration

A cleaned housing dataset is connected to the application to obtain meaningful input options. From this dataset, the system extracts the list of available locations. These locations are then displayed in a dropdown menu so users can select the area where the property is located. This step is important because it prevents invalid inputs and ensures that the model receives data similar to what it was trained on, which improves prediction accuracy.

User Interface Design

The interface is designed to be simple, clean, and easy to understand. The main title and subtitle clearly indicate that the application is a Bangalore Housing Price Estimator. The layout is centered so that all elements appear organized on the screen. Instructions and headings guide the user to enter the required property details. The goal of the design is to provide a smooth user experience without confusion.

User Input Collection

The application collects property information through interactive fields. Users can enter numerical details such as the total area of the house in square feet, the number of bathrooms, and the number of bedrooms (BHK). In addition, users select the property location from a dropdown list of available areas. These inputs represent the key factors that influence house prices and are required by the prediction model.

Prediction Trigger Mechanism

A “Predict Price” button is provided to start the estimation process. This ensures that the model runs only after the user has entered all necessary details. It also gives users control over when they want to generate the prediction. Once clicked, the system begins processing the inputs.

Prediction Process

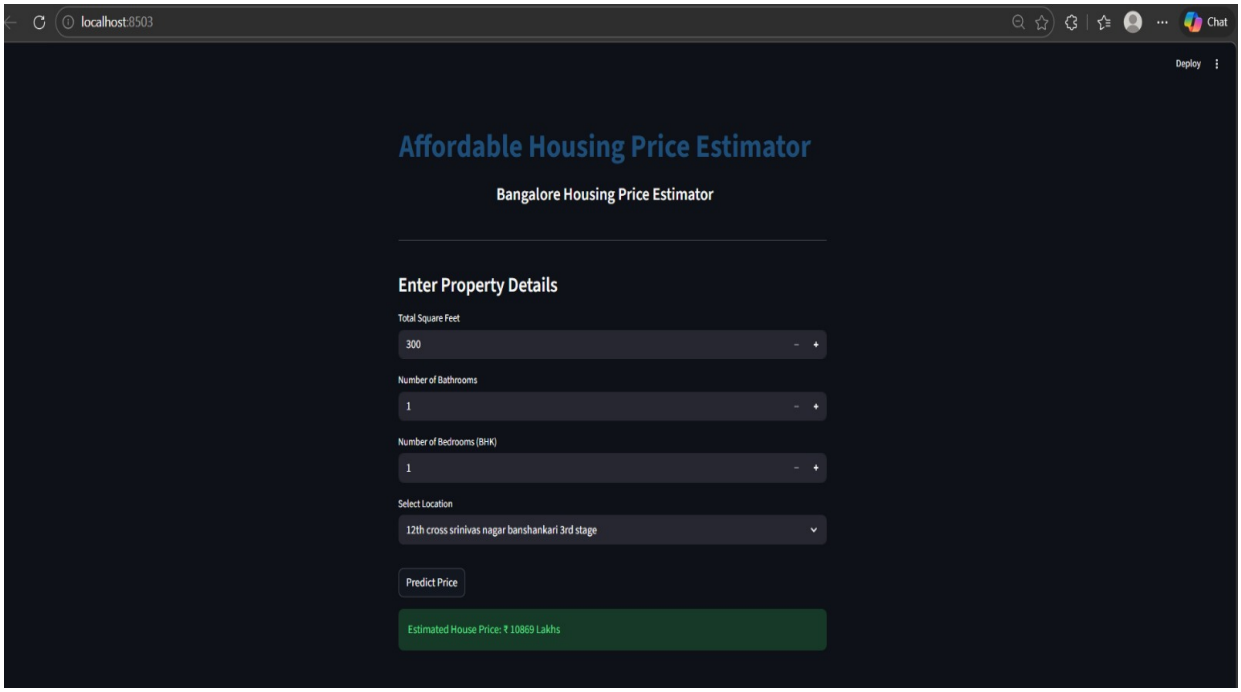
After the button is pressed, the application converts the entered details into the numerical format expected by the trained model. The model then analyzes the inputs using the relationships it learned during training, such as how property size and number of rooms affect price. Based on this analysis, the model calculates an estimated market value for the property.

Result Display

The predicted house price is displayed immediately on the screen in a highlighted message. This instant feedback helps users quickly understand the estimated value of the property they described. The clear presentation of results makes the application practical for quick decision-making.

app.py > ...

```python
import streamlit as st
import pickle
import pandas as pd
import numpy as np


model = pickle.load(open("house_model.pkl", "rb"))
st.write(model.coef_)
st.write("Number of features expected:", len(model.coef_))

data = pd.read_csv("Bengaluru_House_Data_Cleaned.csv")
locations = sorted(data['location'].unique())


st.set_page_config(
    page_title="house_price_projectS",
    layout="centered"
)


st.markdown(
    "<h1 style='text-align:center; color:#1f4e79;'>house_price_projectS</h1>",
    unsafe_allow_html=True
)

st.markdown(
    "<h4 style='text-align:center;'>Bangalore Housing Price Estimator</h4>",
    unsafe_allow_html=True
)

st.write("---")

st.subheader("Enter Property Details")


total_sqft = st.number_input("Total Square Feet", min_value=300)
bath = st.number_input("Number of Bathrooms", min_value=1)
bhk = st.number_input("Number of Bedrooms (BHK)", min_value=1)
location = st.selectbox("Select Location", locations)

st.write("")


if st.button("Predict Price"):

    input_data = np.array([[total_sqft, bath]])  # change if needed

    prediction = model.predict(input_data)

    st.success(f"Estimated House Price: ₹ {round(prediction[0], 2)} Lakhs")
```


localhost:8503
Chat
Deploy
Affordable Housing Price Estimator
Bangalore Housing Price Estimator
Enter Property Details
Total Square Feet
300
Number of Bathrooms
1
Number of Bedrooms (BHK)
1
Select Location
12th cross srinivas nagar banshankari 3rd stage
Predict Price
Estimated House Price: ₹ 10869 Lakhs